

Transport Layer Protocols

TCP and UDP

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Module 4 — what we covered

OUR ROUTE THROUGH THE
TRANSPORT LAYER

Module 5 roadmap

Transport-layer services

Applications use transport services for process-to-process communication.

Transport layer provides an end-to-end channel between applications

Suppose a network application runs process A and process B on separate devices and needs an end-to-end communication channel.



HOW THE ABSTRACTION IS DELIVERED

Mechanisms utilized by transport- layer protocols

To provide this end-to-end abstraction, transport protocols use several mechanisms.

ONE HOST • MANY APPLICATION
PROCESSES

Why multiplexing is needed

COMBINE OUTGOING DATA • ROUTE INCOMING
DATA

Multiplexing sends; demultiplexing delivers

BEFORE DEMULTIPLEXING

Sockets are transport-layer endpoints

UDP demultiplexing

Suppose the network layer uses the destination IP address to deliver the following IP packet to some host.

How does UDP perform demultiplexing afterwards?

ONE IP PACKET CARRYING ONE UDP DATAGRAM

IP HEADER

SOURCE IP
198.51.100.23

DESTINATION IP
192.0.2.80

UDP HEADER

SOURCE PORT
42345

DESTINATION PORT
19157

application data

TCP demultiplexing

Suppose the network layer uses the destination IP address to deliver the following IP packet to some host.

How does TCP perform demultiplexing afterwards?

TCP CONNECTION IDENTITY

Source endpoint

SOURCE IP ADDRESS
203.0.113.10

SOURCE PORT
46428

Destination endpoint

DESTINATION IP ADDRESS
192.0.2.80

UDP and TCP

One is connectionless and minimal; the other is connection-oriented and reliable.

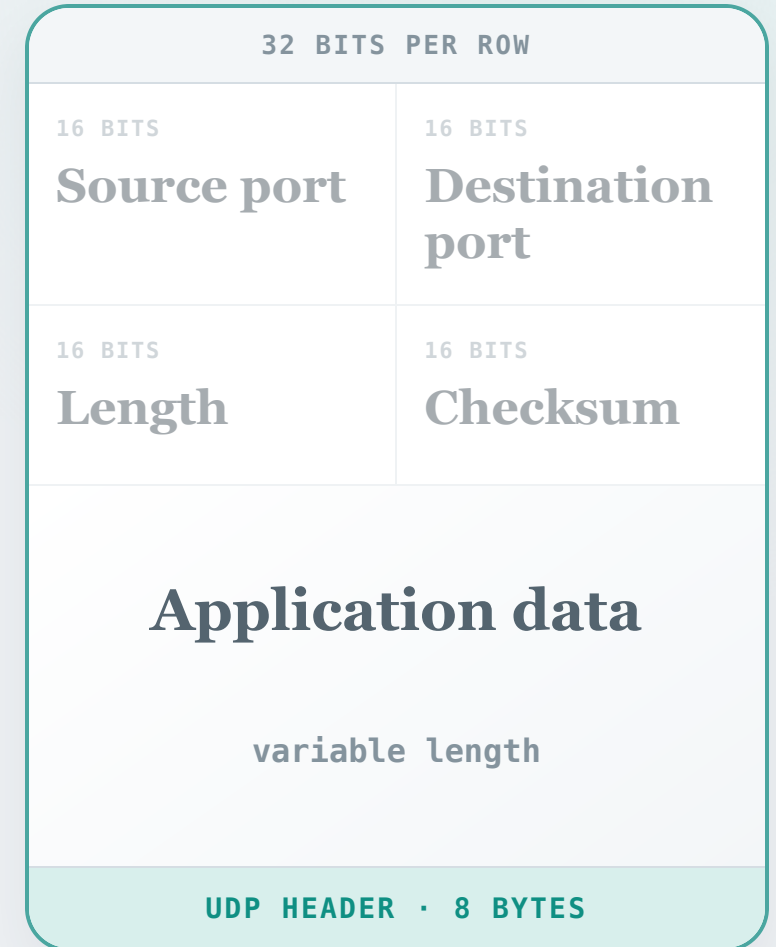
USER DATAGRAM PROTOCOL
• RFC 768

User Datagram Protocol: an overview

UDP is connectionless

FOUR FIELDS • EIGHT-BYTE HEADER

A UDP segment has a compact header



THE BENEFIT OF A MINIMAL
TRANSPORT SERVICE

Why do some applications choose UDP?

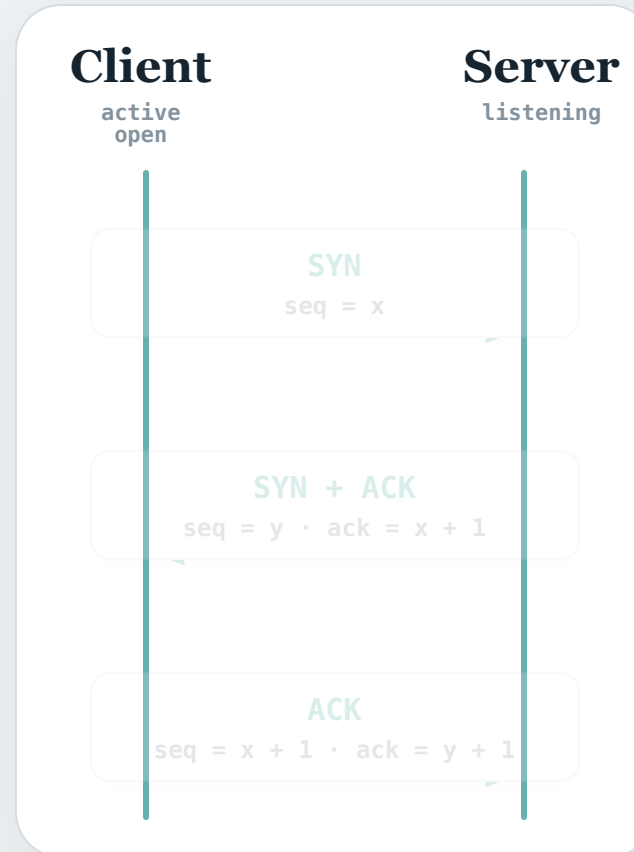
TRANSMISSION CONTROL
PROTOCOL · RFC 9293

TCP: an overview

TCP is connection- oriented

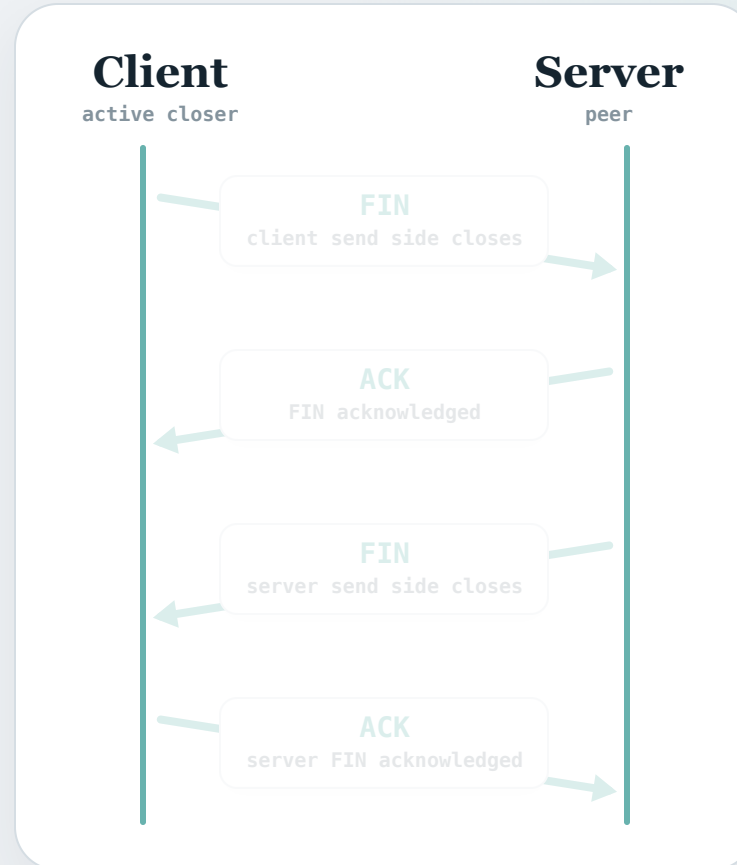
CONNECTION ESTABLISHMENT ·
SYN CONSUMES ONE SEQUENCE
NUMBER

TCP establishes a connection with a three-way handshake



CONNECTION TEARDOWN •
EACH DIRECTION CLOSES
INDEPENDENTLY

How TCP closes a connection



A COMPACT BREAKDOWN OF THE
TCP HEADER

Key TCP header fields



PIPELINING ·
CUMULATIVE
ACKNOWLEDGMENTS ·
RETRANSMISSION

The RDT used by TCP — an overview

BYTE STREAM • EVERY
BYTE HAS ITS OWN
NUMBER

TCP

sequence numbers count bytes

ISN = 1000 · MSS = 1000 BYTES · 3000 BYTES TO SEND

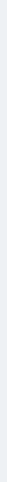
Sequence and acknowledgment in action

The TCP sender rules: overview

TCP fast retransmit

Sender A

Receiver B



TCP receiver rules: overview

TCP timeout: the challenge

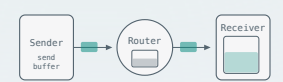
TCP RTT estimation

TCP timeout interval

TCP Flow Control

TCP uses receiver feedback so that a fast sender cannot swamp a slow receiver.

Flow control vs congestion control



SPEED-MATCHING THE SENDER TO THE RECEIVER

TCP flow control: the receive window

FROM RECEIVER TO SENDER

Advertising and enforcing the receive window

Congestion Control

Too many senders overload router buffers. TCP limits each sender's rate to prevent this.

Approaches to congestion control



TCP congestion detection and action: an overview

How TCP matches its sending rate



TCP • slow start

Sender

Receiver

TCP • congestion avoidance

Sender

Receiver

Why AIMD?

TCP • reacting to loss

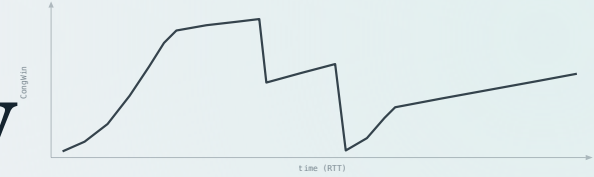
TCP · fast recovery



TCP • fast recovery exit



TCP • congestion control overview



Congestion control — summary

TCP begins in **slow start**, doubling CongWin each RTT until ssthresh, then grows linearly in **congestion avoidance**.

Three dup ACKs trigger **fast recovery** — ssthresh and CongWin halve. A **timeout** drops CongWin to 1 MSS and restarts slow start.

The resulting **AIMD sawtooth** continuously probes for available bandwidth while backing off sharply on congestion signals.

Acknowledgements

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- Prof Aleksandar Kuzmanovic's lecture notes for CS340, Northwestern University, 2005 — https://users.cs.northwestern.edu/~akuzma/classes/CS340-w05/lecture_notes.htm
- Dr Barry Wu's lecture notes for COSC264, University of Canterbury

References

James F. Kurose & Keith W. Ross, *Computer Networking: A Top-Down Approach Featuring the Internet*, Addison Wesley, 3rd edition, 2005.

Chapter 3 Transport Layer (3.1 – 3.3, 3.5 – 3.7)

William Stallings, *Data and Computer Communications*, Pearson, 10th edition, 2013.

Chapter 15 Transport Protocols

Andrew S. Tanenbaum & David J. Wetherall, *Computer Networks*, Prentice Hall, 5th edition, 2010.

Chapter 6 The Transport Layer (6.2, 6.3, 6.5)